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Department of Computer Science & Engineering

LABORATORY FILE

Subject	Full Stack Development
LABSHEET NO.	02
Experiment Title	Making It Interactive (Vanilla JavaScript)
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Experiment No. 02

Title: Making It Interactive (Vanilla JavaScript)

Aim

To add interactivity to the static e-commerce storefront using plain (vanilla) JavaScript — including dynamic product rendering, cart management with localStorage, live total calculation, form validation and a cart-count badge — before moving to a framework such as React.

Theory

Vanilla JavaScript allows direct manipulation of the Document Object Model (DOM) without external libraries. Array methods such as `map()`, `filter()`, `find()` and `reduce()` help transform and summarise data. `localStorage` provides simple client-side persistence so the cart survives page refreshes. Form validation is performed by checking field values and displaying inline error messages. Event listeners (click, change, submit) connect user actions to these functions, creating a fully interactive single-page experience within a multi-page static site.

Software Required

- Visual Studio Code (or any text editor)
- Web Browser (Google Chrome / Microsoft Edge)

Procedure

1. Create a `script.js` file and define an array of at least 8 product objects (id, name, price, image).
2. On `products.html`, replace hardcoded product cards with an empty container and use JavaScript (`map` + `innerHTML`) to render cards dynamically.
3. Implement an Add-to-Cart handler that updates a cart array and saves it to `localStorage`.
4. On `cart.html`, read the cart from `localStorage` and render each item with a quantity input and a Remove button.
5. Write `calculateTotal()` using `reduce()` so the grand total updates whenever quantity changes or an item is removed.
6. On `checkout.html`, validate Name, Address, Pincode (6 digits) and Phone (10 digits) with inline error messages; on success show a confirmation and clear the cart.
7. Add a small script on every page that updates a cart-item-count badge next to the Cart link.

Source Code

1. script.js (complete)

```
/* =====
GadgetHub - Mini E-Commerce Storefront
script.js | Lab Sheet 2 - Vanilla JS
===== */

// ----- 1. Product Catalog (at least 8 products) -----
const products = [
  { id: 1, name: "Wireless Earbuds Pro", price: 2499, image: "https://via.placeholder.com/300x200?text=Wireless+Earbuds" },
  { id: 2, name: "Smart Watch X5", price: 4999, image: "https://via.placeholder.com/300x200?text=Smart+Watch" },
  { id: 3, name: "Portable Bluetooth Speaker", price: 1799, image:
"https://via.placeholder.com/300x200?text=Bluetooth+Speaker" },
  { id: 4, name: "20000mAh Power Bank", price: 1299, image: "https://via.placeholder.com/300x200?text=Power+Bank" },
  { id: 5, name: "7-in-1 USB-C Hub", price: 2199, image: "https://via.placeholder.com/300x200?text=USB-C+Hub" },
  { id: 6, name: "Ergonomic Wireless Mouse", price: 899, image: "https://via.placeholder.com/300x200?text=Wireless+Mouse"
},
  { id: 7, name: "Aluminum Laptop Stand", price: 1499, image: "https://via.placeholder.com/300x200?text=Laptop+Stand" },
  { id: 8, name: "RGB Mechanical Keyboard", price: 3499, image:
"https://via.placeholder.com/300x200?text=Mechanical+Keyboard" }
];

// ----- Cart helpers (localStorage) -----
function getCart() {
  return JSON.parse(localStorage.getItem("cart")) || [];
}

function saveCart(cart) {
  localStorage.setItem("cart", JSON.stringify(cart));
  updateCartBadge();
}

// ----- 8. Cart badge on every page -----
function updateCartBadge() {
  const cart = getCart();
  const count = cart.reduce((sum, item) => sum + item.quantity, 0);
  const badge = document.getElementById("cart-count");
  if (badge) {
    badge.textContent = count;
    badge.style.display = count > 0 ? "inline-block" : "none";
  }
}

// ----- 3. Add to Cart -----
function addToCart(productId) {
  const product = products.find(p => p.id === productId);
  if (!product) return;

  let cart = getCart();
  const existing = cart.find(item => item.id === productId);

  if (existing) {
    existing.quantity += 1;
  } else {
    cart.push({ ...product, quantity: 1 });
  }

  saveCart(cart);
  alert(`"${product.name}" added to cart!`);
}
```

```
// ----- 2. Dynamically render products on products.html -----
function renderProducts() {
  const container = document.getElementById("product-list");
  if (!container) return;

  container.innerHTML = products.map(p => `
    <div class="product-card">
      
      <div class="card-body">
        <h3>${p.name}</h3>
        <p class="price">₹${p.price.toLocaleString("en-IN")}</p>
        <button class="btn" onclick="addToCart(${p.id})">Add to Cart</button>
      </div>
    </div>
  `).join("");
}

// ----- 4 & 5. Cart page - render items + calculateTotal -----
function calculateTotal() {
  const cart = getCart();
  const total = cart.reduce((sum, item) => sum + item.price * item.quantity, 0);
  const totalEl = document.getElementById("grand-total");
  if (totalEl) {
    totalEl.textContent = "₹" + total.toLocaleString("en-IN");
  }
  return total;
}

function updateQuantity(productId, newQty) {
  let cart = getCart();
  const item = cart.find(i => i.id === productId);
  if (!item) return;

  newQty = parseInt(newQty, 10);
  if (newQty < 1) newQty = 1;
  item.quantity = newQty;
  saveCart(cart);
  renderCart();
}

function removeFromCart(productId) {
  let cart = getCart().filter(item => item.id !== productId);
  saveCart(cart);
  renderCart();
}

function renderCart() {
  const tbody = document.getElementById("cart-body");
  if (!tbody) return;

  const cart = getCart();

  if (cart.length === 0) {
    tbody.innerHTML = `<tr><td colspan="5" style="text-align:center;padding:2rem;">Your cart is empty.</td></tr>`;
    calculateTotal();
    return;
  }

  tbody.innerHTML = cart.map(item => `
    <tr>
```

```
        <td>${item.name}</td>
        <td>
            <input type="number" min="1" value="${item.quantity}"
                style="width:60px;padding:4px;text-align:center;"
                onchange="updateQuantity(${item.id}, this.value)">
        </td>
        <td>₹${item.price.toLocaleString("en-IN")}</td>
        <td>₹${(item.price * item.quantity).toLocaleString("en-IN")}</td>
        <td><button class="btn" style="padding:4px 10px;font-size:0.85rem;background:#d93025;"
            onclick="removeFromCart(${item.id})">Remove</button></td>
    </tr>
` ).join("");

    calculateTotal();
}

// ----- 6 & 7. Checkout form validation + success -----
function validateCheckout(event) {
    event.preventDefault();

    // Clear previous errors
    document.querySelectorAll(".error-msg").forEach(el => el.textContent = "");

    const name = document.getElementById("name").value.trim();
    const address = document.getElementById("address").value.trim();
    const city = document.getElementById("city").value.trim();
    const pincode = document.getElementById("pincode").value.trim();
    const phone = document.getElementById("phone").value.trim();

    let valid = true;

    if (name === "") {
        document.getElementById("err-name").textContent = "Name is required.";
        valid = false;
    }
    if (address === "") {
        document.getElementById("err-address").textContent = "Address is required.";
        valid = false;
    }
    if (city === "") {
        document.getElementById("err-city").textContent = "City is required.";
        valid = false;
    }
    if (!/^\d{6}$/.test(pincode)) {
        document.getElementById("err-pincode").textContent = "Pincode must be exactly 6 digits.";
        valid = false;
    }
    if (!/^\d{10}$/.test(phone)) {
        document.getElementById("err-phone").textContent = "Phone must be exactly 10 digits.";
        valid = false;
    }
}

if (!valid) return;

// Success
localStorage.removeItem("cart");
updateCartBadge();

const form = document.getElementById("checkout-form");
const msg = document.getElementById("success-msg");
if (form) form.style.display = "none";
if (msg) {
```

```
    msg.style.display = "block";
    msg.textContent = "Order placed successfully! Thank you for shopping with GadgetHub.";
  }
}

// ----- Init on page load -----
document.addEventListener("DOMContentLoaded", () => {
  updateCartBadge();
  renderProducts(); // only runs if #product-list exists
  renderCart();     // only runs if #cart-body exists

  const form = document.getElementById("checkout-form");
  if (form) {
    form.addEventListener("submit", validateCheckout);
  }
});
```

2. products.html (updated – dynamic rendering)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>GadgetHub | Products</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <nav class="navbar">
      <a href="index.html" class="logo">Gadget<span>Hub</span></a>
      <ul class="nav-links">
        <li><a href="index.html">Home</a></li>
        <li><a href="products.html" class="active">Products</a></li>
        <li>
          <a href="cart.html">Cart
            <span id="cart-count" style="display:none;background:#ff6d00;color:#fff;border-radius:50%;padding:1px 7px;font-size:0.75rem;margin-left:4px;">0</span>
          </a>
        </li>
      </ul>
    </nav>
  </header>

  <main>
    <h2 class="section-title">All Products</h2>
    <!-- Product cards injected by JavaScript -->
    <div class="product-grid" id="product-list"></div>
  </main>

  <footer>
    <p><strong>GadgetHub</strong> – Your Trusted Tech Store</p>
    <p>Email: support@gadgethub.com | Phone: +91 98765 43210</p>
    <p>&copy; 2026 GadgetHub. All rights reserved.</p>
  </footer>

  <script src="script.js"></script>
</body>
</html>
```

3. cart.html (updated – dynamic cart)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>GadgetHub | Cart</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <nav class="navbar">
      <a href="index.html" class="logo">Gadget<span>Hub</span></a>
      <ul class="nav-links">
        <li><a href="index.html">Home</a></li>
        <li><a href="products.html">Products</a></li>
        <li>
          <a href="cart.html" class="active">Cart
```

```

        <span id="cart-count" style="display:none;background:#ff6d00;color:#fff;border-radius:50%;padding:1px 7px;font-size:0.75rem;margin-left:4px;">0</span>
      </a>
    </li>
  </ul>
</nav>
</header>

<main>
  <h2 class="section-title">Your Shopping Cart</h2>
  <table class="cart-table">
    <thead>
      <tr>
        <th>Item Name</th>
        <th>Quantity</th>
        <th>Price</th>
        <th>Subtotal</th>
        <th>Action</th>
      </tr>
    </thead>
    <tbody id="cart-body">
      <!-- Cart items rendered by JavaScript -->
    </tbody>
    <tfoot>
      <tr class="total-row">
        <td colspan="3">Grand Total</td>
        <td id="grand-total" colspan="2">₹0</td>
      </tr>
    </tfoot>
  </table>

  <div class="cart-actions">
    <a href="products.html" class="btn" style="background:#5f6368;">Continue Shopping</a>
    <a href="checkout.html" class="btn">Proceed to Checkout</a>
  </div>
</main>

<footer>
  <p><strong>GadgetHub</strong> – Your Trusted Tech Store</p>
  <p>Email: support@gadgethub.com | Phone: +91 98765 43210</p>
  <p>&copy; 2026 GadgetHub. All rights reserved.</p>
</footer>

  <script src="script.js"></script>
</body>
</html>

```

4. checkout.html (updated – form validation)

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>GadgetHub | Checkout</title>
  <link rel="stylesheet" href="style.css">
  <style>
    .error-msg { color: #d93025; font-size: 0.85rem; margin-top: 4px; min-height: 18px; }
    #success-msg {
      display: none;
      background: #e6f4ea;
      color: #137333;
      padding: 1.5rem;
      border-radius: 10px;
    }
  </style>

```

```
    text-align: center;
    font-size: 1.15rem;
    font-weight: 600;
    max-width: 600px;
    margin: 0 auto;
  }
</style>
</head>
<body>
  <header>
    <nav class="navbar">
      <a href="index.html" class="logo">Gadget<span>Hub</span></a>
      <ul class="nav-links">
        <li><a href="index.html">Home</a></li>
        <li><a href="products.html">Products</a></li>
        <li>
          <a href="cart.html">Cart
            <span id="cart-count" style="display:none;background:#ff6d00;color:#fff;border-radius:50%;padding:1px 7px;font-size:0.75rem;margin-left:4px;">0</span>
          </a>
        </li>
      </ul>
    </nav>
  </header>

  <main>
    <h2 class="section-title">Checkout</h2>

    <div id="success-msg"></div>

    <form class="checkout-form" id="checkout-form">
      <div class="form-group">
        <label for="name">Full Name</label>
        <input type="text" id="name" name="name" placeholder="Enter your full name">
        <div class="error-msg" id="err-name"></div>
      </div>

      <div class="form-group">
        <label for="address">Address</label>
        <input type="text" id="address" name="address" placeholder="House No., Street, Area">
        <div class="error-msg" id="err-address"></div>
      </div>

      <div class="form-group">
        <label for="city">City</label>
        <input type="text" id="city" name="city" placeholder="Enter city">
        <div class="error-msg" id="err-city"></div>
      </div>

      <div class="form-group">
        <label for="pincode">Pincode</label>
        <input type="text" id="pincode" name="pincode" placeholder="6-digit pincode">
        <div class="error-msg" id="err-pincode"></div>
      </div>

      <div class="form-group">
        <label for="phone">Phone Number</label>
        <input type="tel" id="phone" name="phone" placeholder="10-digit mobile number">
        <div class="error-msg" id="err-phone"></div>
      </div>

      <div class="form-group">
```

```

<label>Payment Method</label>
<div class="radio-group">
  <label><input type="radio" name="payment" value="cod" checked> Cash on Delivery</label>
  <label><input type="radio" name="payment" value="upi"> UPI</label>
  <label><input type="radio" name="payment" value="card"> Credit / Debit Card</label>
</div>
</div>

<button type="submit" class="btn">Place Order</button>
</form>
</main>

<footer>
  <p><strong>GadgetHub</strong> – Your Trusted Tech Store</p>
  <p>Email: support@gadgethub.com | Phone: +91 98765 43210</p>
  <p>&copy; 2026 GadgetHub. All rights reserved.</p>
</footer>

<script src="script.js"></script>
</body>
</html>

```

Output

The screenshot below show the interactive pages after script.js is linked and the page is opened in a browser.

The screenshots show the following pages of the MyShop application:

- Products Page:** Displays a grid of products including Wireless Headphones, Smart Watch, Backpack, Sneakers, Sunglasses, Bluetooth Speaker, Hoodie, and Water Bottle, each with an 'Add to Cart' button.
- Your Cart Page:** Shows a table of items in the cart:

Product	Price	Quantity	Total	Actions
Wireless Headphones	₹999	1	₹999	Remove
Smart Watch	₹2499	2	₹4998	Remove
Bluetooth Speaker	₹1499	1	₹1499	Remove
Grand Total:			₹7496	

 A 'Proceed to Checkout' button is at the bottom.
- Checkout Page:** A form with fields for Full Name, Address, Pincode, and Phone Number. The 'Place Order' button is at the bottom.
- Order Placed Page:** Displays a green checkmark, the text 'Order placed!', and a message: 'Thank you for your order. Your order has been placed successfully.' A 'Continue Shopping' button is at the bottom.